

DATA CARD

Dataset Documentation · FIAR / CIIA UFMG

Dataset name:

Hospitalizations for Respiratory Diseases

SECTION 0

DATASET SUMMARY

Name	SIHSUS - Hospitalizations for Respiratory Diseases (SUSHRD)
Description	Dataset for hospitalization by respiratory diseases, grouped by hospital and months. Used in forecasting models for predicting hospitalizations based on aggregated data from previous months.
Link / Repository	https://github.com/niar-saude-ufmg/SBCAS_26_Respiratory_Disease/blob/main/SIH_Data/Final_Table.zip
Availability	Public — released as the SBCAS publication dataset for the FIAR framework, containing only hospital-aggregated, non-identifiable data.
Data card authors	· Researcher 1, Manager · Researcher 2, Contributor
Release Date	26/02/2026
Current version	2.0

SECTION 1

AUTHORSHIP

1.1 Dataset Owners

OBLIGATORY

Team / Laboratory	NIAR-Saúde
Team's contact	niar@dcc.ufmg.br
Authors	· Researcher 1 · Researcher 2

1.2 Publishers

OBLIGATORY

Institution(s)	CIIA UFMG
Type	Academic

Contact	iasaudeufmg@gmail.com
Website	https://ciia-saude.dcc.ufmg.br/

1.3 Funders

OPTIONAL

Institutions(s)	Not applicable — no external funding tied specifically to this dataset.
Program / Grant	Not applicable
Funding Summary	No external funding tied specifically to this dataset

SECTION 2

DATASET OVERVIEW

2.1 Overall Features

OBLIGATORY

Data Type	Time series
File Format	CSV
Total Size	210.2 MB
Total Instances	161,584 instances
Total Features	141 features
Covered Time Frame	01/2022 – 11/2025
Geographical coverage	National

2.2 Content Description

OBLIGATORY

Instance Description	Each line holds hospital accumulated statistics for a given month and past months.
Features Description	Features include hospital characteristics, time frame, and for previous and current months: total hospitalizations, average days spent on the hospital, death ratio, total money spent on the hospital and UTI, ratio of patients belonging to a sensitive group (such as sex or race), ratio of specific respiratory diseases, and regionality coverage.

SECTION 3

SENSITIVITY AND PRIVACY

⚠ Health Data are sensitive data by LGPD (Art. 11) terms. Fill this section with attention.

3.1 Data Classification

OBLIGATORY

Data Origin	Data from SUS (DATASUS, SIH)
Is there personal data?	Yes, anonymized
Types of S/PII present	Sex, Race, Age, ICD-10
Child Data (<18)?	Yes

3.2 Fields with sensitive data

OBLIGATORY

Fields	Type of sensitive data	Collected intentionally?	Protective measure applied
J_race_white_share_lagX J_race_black_share_lagX J_race_others_share_lagX	Demographic data — race/color	Yes	Aggregated data makes it difficult to extract single individuals
J00_06_share_lagX J09_18_share_lagX J20_22_share_lagX J40_47_share_lagX	Health data — diagnosis (ICD-10 group)	Yes	Aggregated data makes it difficult to extract single individuals
J_sex_m_share_lagX J_sex_f_share_lagX	Demographic data — biological sex (technically not sensitive data, but will be used for fairness evaluation)	Yes	Aggregated data makes it difficult to extract single individuals
J_age_60p_share_lagX J_age_15_59_share_lagX J_age_0_14_share_lagX	Demographic data — age group (technically not sensitive data, but will be used for fairness evaluation)	Yes	Aggregated data makes it difficult to extract single individuals

3.3 Legal Basis and Conformity

OBLIGATORY

Legal Basis (LGPD)	Art. 11, II, “c”, of LGPD - processing grounded on studies realized by research institutions, category in which FIAR-Saúde/CIIA-UFGM is included.
Report / CAAE	Not applicable. The research team did not have access to non-anonymized or individually identifiable data at any stage; this dataset is already aggregated by hospital (Section 2.2), which does not constitute human-subjects research requiring CEP/CONEP submission.
TCLE / Waiver	Not applicable. The data used is already aggregated by hospital and anonymized at the source (DATASUS/SIH-SUS), with no access to individual records at any stage of the pipeline — therefore, there is no direct collection from data subjects that would require informed consent (TCLE). [Confirm with UFGM's DPO whether a formal CEP waiver is required for this secondary use of data]
Sharing agreement	No formal DPA (Data Processing Agreement) or partnership agreement with the data source. Data was accessed via FTP download from DATASUS/SIH-SUS under its public open-data terms of use (see Section 4.1). The resulting aggregated dataset was subsequently published in a public GitHub repository as part of the SBCAS publication, independent of any agreement with DATASUS..
Security measures	Data is publicly available in the niar-saude-ufmg GitHub repository. Given the public nature of this release, protection relies on the irreversibility of the aggregation applied upstream (hospital-level statistics, no individual records) rather than access control. No additional identifiers are introduced at any processing stage.

3.4 Risks and Mitigation

OBLIGATORY

Type(s) of Risk	Indirect Risk (crossing with other datasets)
Risk Description	Since all data is aggregated (there is no data on an individual level), and low monthly-average hospitals were filtered out, the data forms a mesh of different people, making it difficult to trace information back to a single person. Still, crossing this data with the original dataset from DataSUS, it may be possible to identify individuals. As the dataset is publicly available (see Section 0), this residual risk is not access-controlled — mitigation relies solely on the irreversibility of the aggregation and the exclusion of low-volume hospitals (Section 4.2), which reduces the granularity available for cross-referencing.
Links de documentação	No dedicated Security POP, DPIA, or formal access policy document exists for this project at this stage. The closest related artifact is the Fairness Report, which analyzes the consequences of the biases identified in this dataset (see Section 6.2) — though it does not cover security or risk-type documentation specifically.

SECTION 4

PROVENANCE AND PROCESSING

4.1 Sources

OBLIGATORY

Source(s)	DATASUS / SIH-SUS
Collection method	FTP connection and downloading
Collection Cadence	Static (single collection)
Time Frame	02/2026
Is the source sensitive?	Yes, since every instance in the original data is the hospitalization of an individual, with their characteristics and personal data.

4.2 Inclusion and Exclusion Criteria

OBLIGATORY

Inclusion Criteria	Diagnosis of ICD-10 starting with J (respiratory diseases).
Exclusion Criteria	Hospitals with lower average than 5 monthly hospitalizations for respiratory diseases.

4.3 Data Processing

OBLIGATORY

Processing Summary	Data was first converted to ".csv" files. Then it was filtered based on columns of interest. Later, new variables were created, aggregating previous collected data. Lastly, the dataset was created by adding new columns, containing information of previous months.
--------------------	--

Used Tools	Python scripts (or other languages).
Source Integration	Not applicable — single data source (DATASUS/SIH-SUS); no linkage across sources was required.
Descriptive Document	https://github.com/nlar-saude-ufmg/SBCAS_26_Respiratory_Disease/v2.ipynb (processing pipeline documented in the project notebook)

SECTION 5

MOTIVATION AND INTENDED USE

5.1 Motivation

OBLIGATORY

Why was this dataset created?	Test application of the FIAR workflow on an experimental model. This dataset was created to train this model.
Application Domain	Hospital Management
FIAR project goal	This dataset is used for training an experimental model to test the FIAR workflow and its capabilities.

5.2 Intended uses and Restrictions

OBLIGATORY

Intended use cases	Prediction model training for expected number of admission; Epidemiologic analysis of respiratory diseases
Unintended use cases	Individual predictions; Unauthorized use for commercial ends; Unsupervised medical decision making; Unrestricted public sharing of the dataset.
Use restrictions	Restricted to FIAR researchers
Citation guideline	@dataset{test, author = {NIAR-Saúde}, title = {Hospitalizations for Respiratory Diseases}, year = {2026}, publisher = {CIIA-UFMG}, }

SECTION 6

HUMAN AND SENSITIVE FEATURES

6.1 Represented Features

OBLIGATORY

Present Features	Biological sex Race / Color (Self-declared) Age / Age group Municipality / State of residency Diagnosys (ICD/CIAP)
Inclusion motivation	These features were included to increase model performance and for Fairness evaluation.

Collection method	Mixed: Extracted from official registries and algorithmically inferred (age groups)
-------------------	---

6.2 Risk of Bias and Mitigation

OBLIGATORY

Identified risk of bias	Severe underrepresentation of black people, and mild underrepresentation of 15-59 year old people and North based hospitals.
Known correlations	No known correlation.
Implemented mitigations	No mitigation was done in the scope of this dataset, as the impact of the underrepresentation is unknown. Mitigations were adopted on a model basis; more can be found in the Fairness Report.

SECTION 7

MAINTENANCE AND VERSION

7.1 Status and Current Version

OBLIGATORY

Maintenance status	Limited maintenance
Current Version	2.0
Release Date	02/2026
Last Updated	02/2026
Maintenance Plan	A new version (minor patch) will only happen if a major error is found.

SECTION 8

TERMS OF ART

Write about acronyms, technical terms and concepts used in this Data card. Use standard definitions when possible.

Term / Acronym	Definition	Source / Reference
SUS	Sistema Único de Saúde (Single Health System)	https://www.gov.br/saude/pt-br/sus
SIH	Sistema de Informações Hospitalares (Hospital Information System)	http://sihd.datasus.gov.br/
PII	Personally Identifiable Information	https://www.machertecnologia.com.br/lgpd-pii-personally-identifiable-information/
LGPD	Lei Geral de Proteção de Dados (General Data Protection Law)	https://www.gov.br/mds/pt-br/acesso-a-informacao/privacidade-e-protecao-de-dados/lgpd
ICD-10	International Classification of Diseases (10th revision)	https://icd.who.int/browse10/
CAAE	Certificado de Apresentação para Avaliação Ética (Presentation for Ethical Assessment)	https://biblioteca.einstein.br/certificado-de-apresentacao-para-apreciacao-etica-caae

TCLE	Termo de Consentimento Livre e Esclarecido (Informed Consent Form)	https://www.gov.br/hubrasil/pt-br/hospitais-universitarios/regiao-sul/husm-ufsm/ensino-e-pesquisa/Modelo2026orientacoesdoTCLE.docx
------	--	---

SECTION 9

REFLECTIONS ON DATA

Use this space to register observations that don't fit in previous sections: known limitations, controversial ethics decisions, or other relevant information for future users of this dataset.

Observation 1	In the repository, all scripts and the dataset may be found under the “./SIH_Data” directory.
Observation 2	The dataset contains data from the year 2022, when the world was still in the covid pandemic. There is a possibility for this data to be skewed.
Observation 3	Due to the amount of data released by DataSUS, processing scripts may take hours to conclude.
Observation 4	This dataset currently lacks dedicated documentation for security measures and risk-type classification (e.g., a Security POP or DPIA). The Fairness Report is the closest related artifact, but it addresses bias consequences rather than security/privacy risk. This gap should be addressed before any transition beyond the research/experimental context.